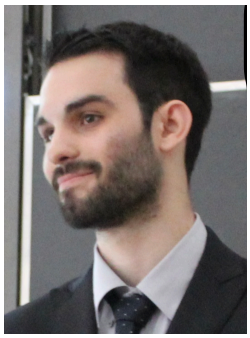
Curriculum Vitae

Mattia Puddu

Date and place of birth: January 29, 1996 – Nuoro, Italy
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Research interests

My current research focuses on extending the theory of Thomas decomposition from algebraic and differential systems of polynomial equations and inequations to the difference-algebraic setting.
More broadly, within difference and differential algebra, I am also interested in the Galois theory of linear difference and differential equations.
Beyond these topics, my mathematical interests include modular forms, classical and difference algebraic geometry, and representation theory, particularly its connections with these areas.

Employment

2025 – **PhD Student in Mathematics**, RWTH Aachen University, Germany
Project: Effective Difference Algebra and Difference Algebraic Groups
Advisors: Prof. Daniel Robertz and Priv.-Doz. Dr. Annette Bachmayr

Preprints

1. *On the geometric degree of the prolongation varieties of the differential system* $x_1' = x_2^2$, $x_2' = x_1^2$ (2026). **Available at:** arXiv:2611.06638.

Code

Difference-Differential-Algebra
Julia/OSCAR implementation of algorithms for algebraic, differential and difference algebra, including Thomas decomposition.

Modular-Forms
Computations involving modular forms using OSCAR, accompanied by mathematical notes.

Education

2019 – 2025 University of Pisa, **Master's Degree in Mathematics**.
Final grade: 110/110 *summa cum laude*
Thesis: *The Differential Galois Group of the Family of All Regular Singular Differential Equations over* $\mathbb{C}(z)$.

Advisor: Prof. Tamás Szamuely
Examiner: Prof. Andrea Maffei
Available at: https://mattiapuddu25.github.io/MasterThesis.html

2015 – 2019 University of Pisa, **Bachelor's Degree in Mathematics**.
Final grade: 110/110 *summa cum laude*.
Thesis: *The Picard-Vessiot Theory – An Introduction with Applications to Fuchsian Differential Equations.*

Advisor: Prof. Paolo Acquistapace.
Available at: https://mattiapuddu25.github.io/BachelorThesis.html

2010 – 2015 Scientific High School "Enrico Fermi", Nuoro, Italy.
Final grade: 100/100 *summa cum laude*.

Conferences, Workshops and Summer Schools

July 2026 FoCM 2026 — Foundations of Computational Mathematics, Workshop on Symbolic Analysis.
Faculty of Mathematics, University of Vienna, Austria, July 16–18, 2026.

Poster: Normal form algorithm for polynomial membership and quasi-simple difference systems.

May 2026 DART 2026 — Differential Algebra and Related Topics XIV.
CNRS Center La Vieille Perrotine, Île d'Oléron, France, May 24–30, 2026.

Poster: Normal form algorithm for polynomial membership and quasi-simple difference systems.

Sep 2025 Summer School "Computer Algebra with OSCAR".
Pfalzakademie, Lambrecht, Germany, September 15–19, 2025.

Teaching experience

2023 University of Pisa — Department of Civil and Industrial Engineering
Tutor for the pre-course *Matematica 0* (Prof. Irene Venturi)
- Held weekly online sessions to help students pass the OFA admission tests

2022 University of Pisa — Department of Agricultural, Food and Agro-Environmental Sciences
Teaching assistant for *Matematica e Statistica* (Bachelor's in Viticulture and Oenology, Prof. Valentino Magnani)
- Conducted the "Mathematics 0" pre-course, including preparation of exams and corresponding solutions
- Held weekly in-class meetings with students for problem-solving and theoretical clarifications
- Reviewed exam papers and solutions from midterm tests throughout the semester
- Supervised students during the exams mentioned above

2021 – 2022 University of Pisa — Department of Computer Science
Tutor for *Algebra Lineare B* (Bachelor's in Computer Science, Prof. Mauro Di Nasso)
- Held meetings with students (mostly upon request) for problem-solving and theoretical clarifications

2019 – 2020 University of Pisa — Department of Physics
Teaching assistant for *Geometria A* (Bachelor's in Physics, Profs. Mario Salvetti and Filippo Di Santo)
- Held meetings with students (mostly upon request) for problem-solving and theoretical clarifications
- Corrected midterm and final exams, including the preparation of written solutions

Computer Skills

Operating systems: Windows, macOS, Linux, Android
Software/languages: MATLAB, C, Julia, OSCAR, HTML (basic), LaTeX (expert)

Language Skills

Italian: native

English: fluent

Spanish: independent user

Mathematical Notes and Resources

Mathematical Notes

Over the years, I have written and collected notes on various mathematical topics, primarily based on courses attended during my university studies in Pisa. Some of these notes have been expanded with additional material reflecting my mathematical interests. They are available on my website, together with brief descriptions of their contents.

Available at: Mathematical Notes

LaTeX Templates

I have developed a collection of LaTeX templates for books, articles, and Beamer presentations, designed to simplify the preparation of mathematical documents. The templates are freely available on my website.

Available at: LaTeX Templates

Math Olympiads

I have compiled two collections of solutions to the Italian Mathematical Olympiad National Team Competitions held in Cesenatico. These collections also include theoretical notes intended to support competition preparation.

Available at: Math Olympiads